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(54) **MUTUAL SECURE COMMUNICATIONS**

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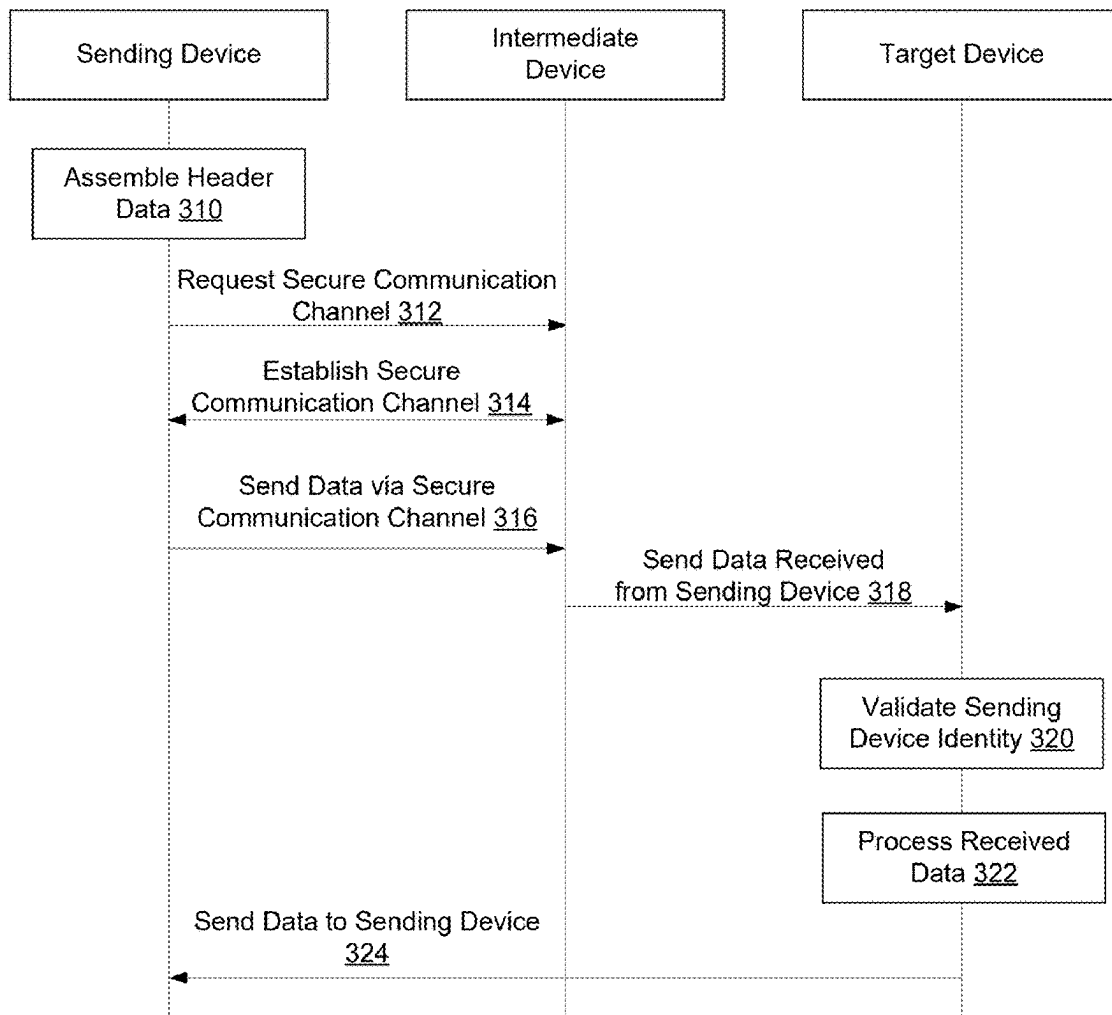
(22) Filed: **Jan. 21, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/478,419, filed on Sep. 29, 2023, now Pat. No. 12,238,095, which is a continuation of application No. 17/883,162, filed on Aug. 8, 2022, now Pat. No. 11,824,853, which is a continuation of application No. 16/376,162, filed on Apr. 5, 2019, now Pat. No. 11,457,010.

(57) **ABSTRACT**

A sending device may send data intended for a target device. An intermediate device may intercept the data sent from the sending device and forward the communications to the target device. Security data (e.g., a security certificate for authentication) along with an encrypted version of the security data may be sent at the application layer such that it passes from the sending device, through the intermediate device, and to the target device without being analyzed or modified by the intermediate device. The target device may use the encrypted security data and the security data to verify the identity of the sending device.



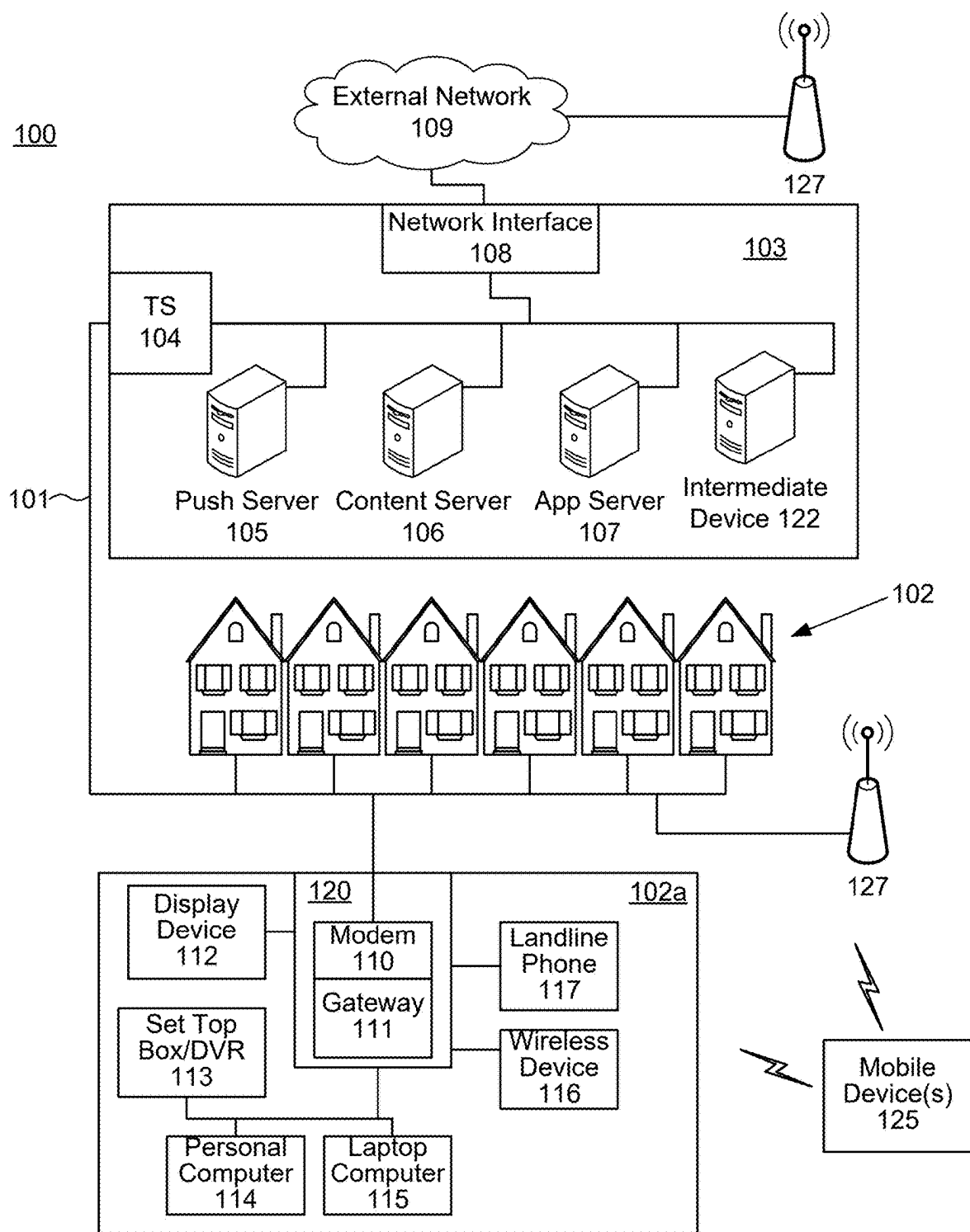


FIG. 1

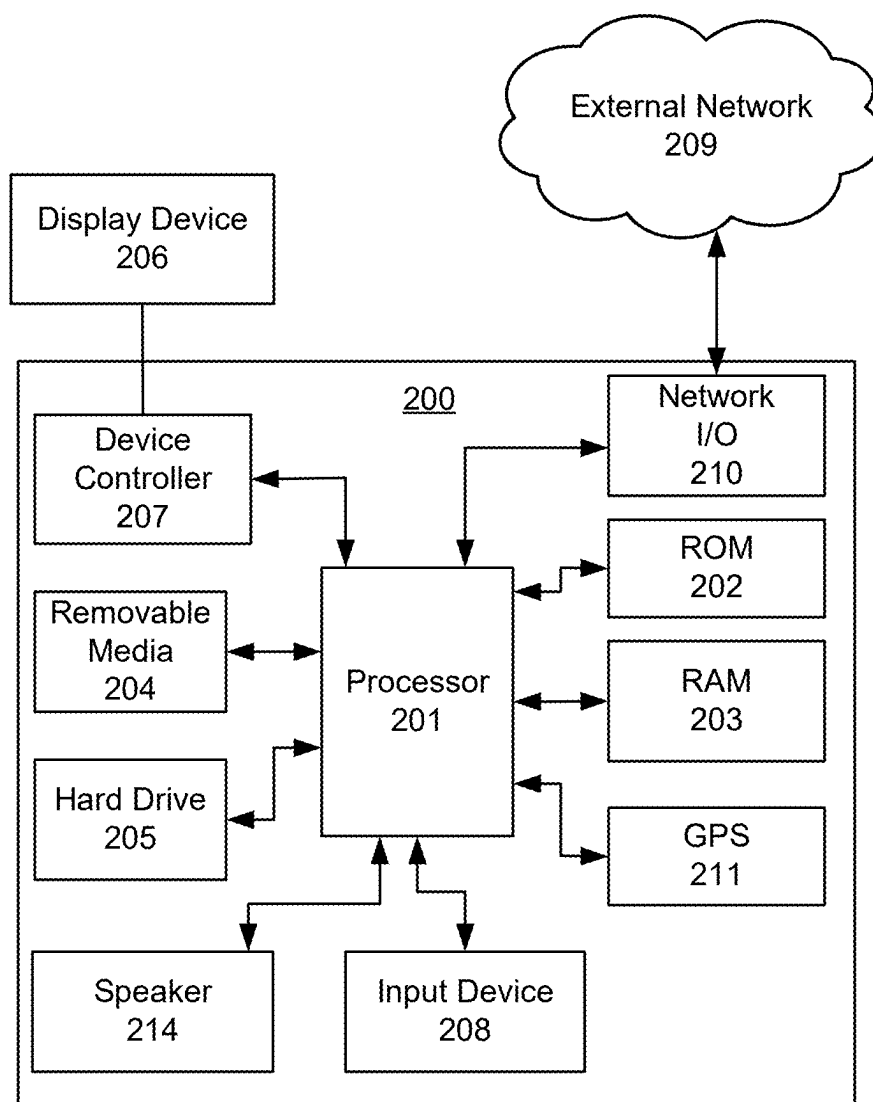


FIG. 2

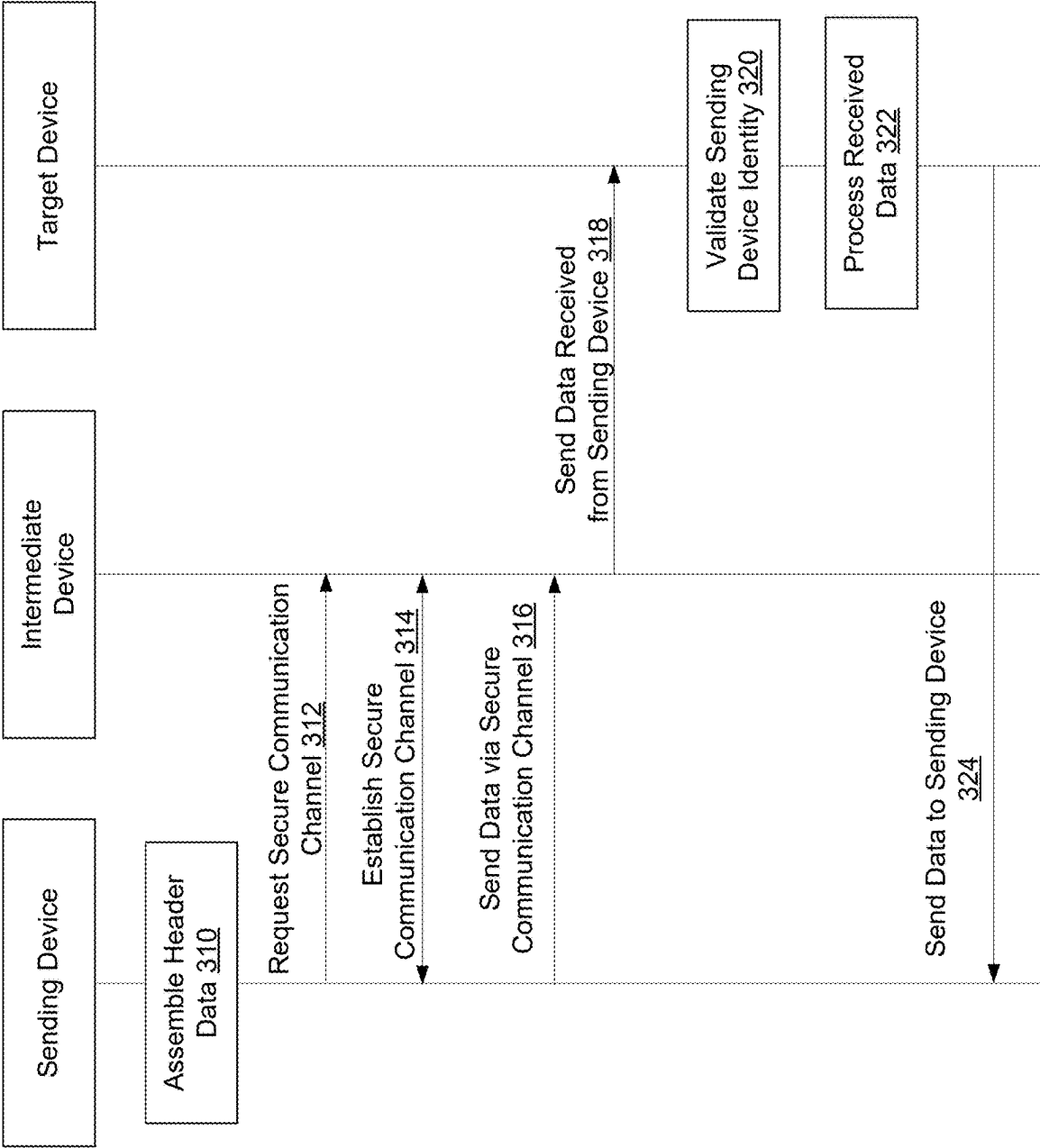


FIG. 3

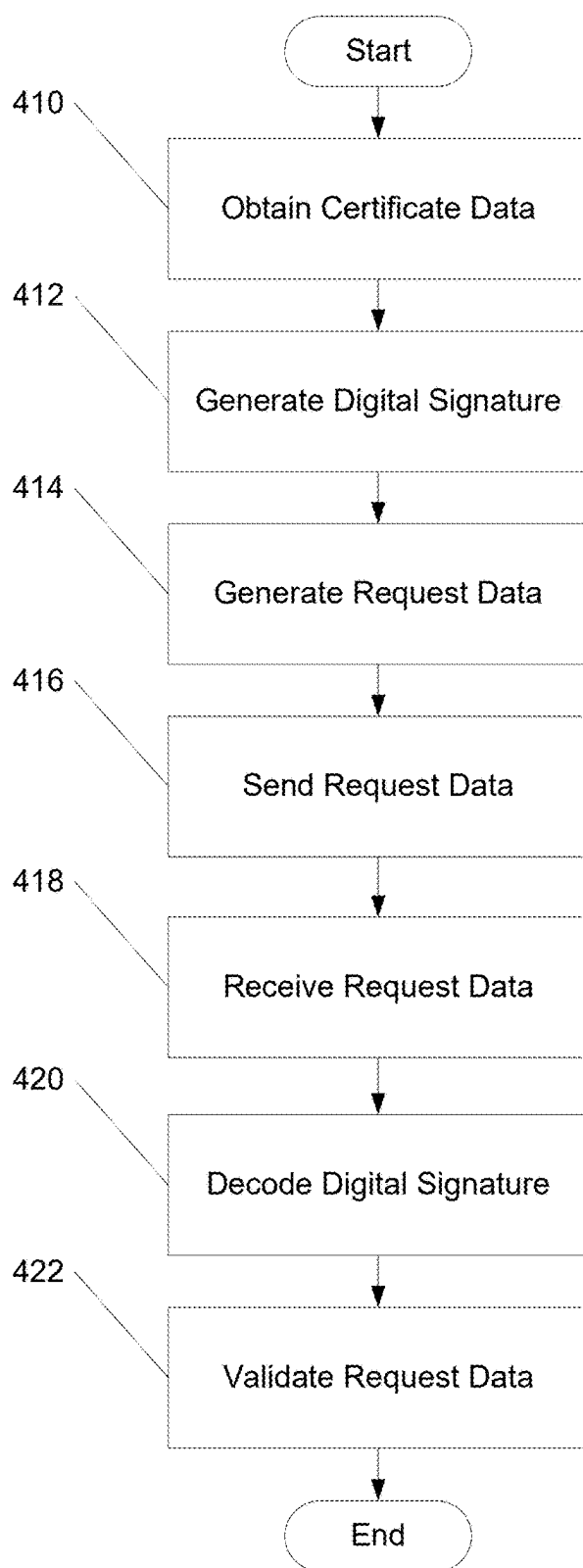


FIG. 4

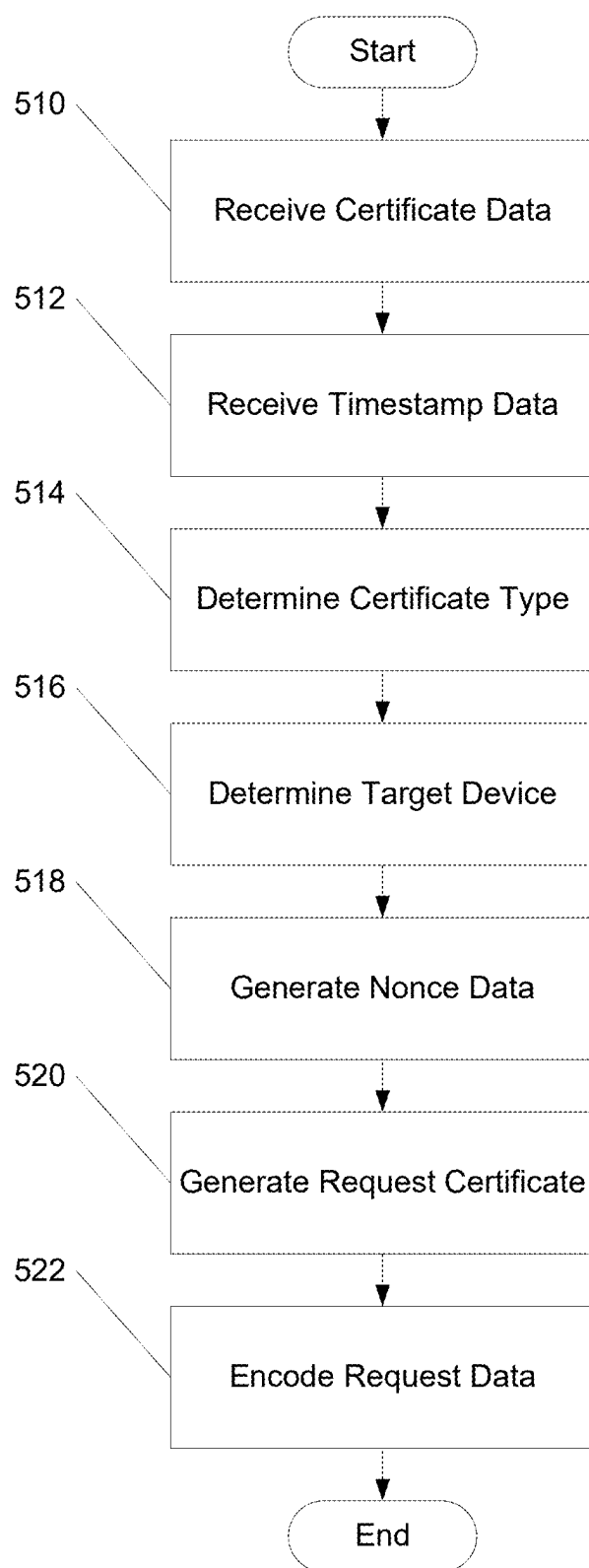


FIG. 5

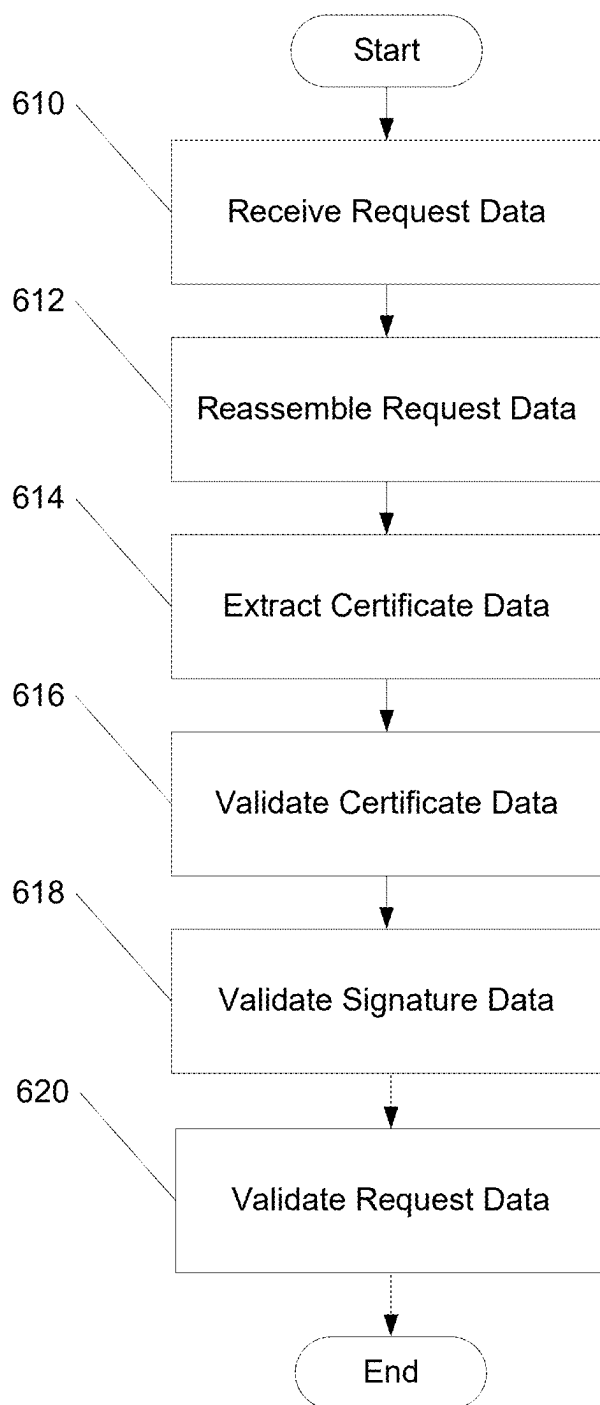


FIG. 6

MUTUAL SECURE COMMUNICATIONS

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of and claims priority to U.S. patent application Ser. No. 18/478,419, filed Sep. 29, 2023, which is a continuation of U.S. patent application Ser. No. 17/883,162, filed Aug. 8, 2022 (now U.S. Pat. No. 11,824,853), which is a continuation of U.S. patent application Ser. No. 16/376,162, filed Apr. 5, 2019 (now U.S. Pat. No. 11,457,010), each of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] Transport Layer Security (TLS) is a standard security protocol (RFC 5246) providing a cryptographic solution for a client (such as a web browser) to authenticate a server. TLS may establish an encrypted secure channel over which the client and server can exchange information. Typical security protocols, such as TLS, are designed to operate between a client and server as end points for a secure communications channel. However, there are situations in which a client communicates with an intermediate device, such as a gateway or load balancer, rather than directly with the server, and the intermediate device may present a vulnerability under the traditional TLS protocol, if the intermediate device were to tamper with security data being exchanged via TLS. These and other shortcomings are addressed by this disclosure.

SUMMARY

[0003] The following summary presents a simplified summary of certain features. The summary is not an extensive overview and is not intended to identify key or critical elements.

[0004] Systems, apparatuses, and methods are described that allows a sending device and target device to authenticate each other in the presence of an intermediate device. The intermediate device may receive incoming security protocol-related data and messages, perform intermediate-type authentication of a computing device, and/or establish a secure communication channel with the sending device. The intermediate device may receive incoming communications and route those communications to a variety of computing devices. The intermediate device may decrypt encrypted data sent by the sending device, and pass the unencrypted data to the target device. Security data (e.g., a security certificate for authentication) may be structured within the data sent by the sending device at an application layer such that it passes through the intermediate device without the security data being extracted or processed by the intermediate device. For example, a security protocol message sent from the sending device and to the target device may include two forms of an authentication certificate issued to the target device—one that is unencrypted, and another that is encrypted using a key associated with the target device. The intermediate device may use the unencrypted form for compatibility with security protocols, but the encrypted form may be inaccessible to the intermediate device, and may be passed on to the target device undisturbed, to allow the target device to determine whether the intermediate device has tampered with the security protocol messages or certificate. The target device can use the security data to authenticate the identity

of the sending device outside of the secure communication channel without the intermediate device having access to and/or decoding the security data.

[0005] These and other features and advantages are described in greater detail below.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Some features are shown by way of example, and not by limitation, in the accompanying drawings. In the drawings, like numerals reference similar elements.

[0007] FIG. 1 shows an example communication network.

[0008] FIG. 2 shows hardware elements of a computing device.

[0009] FIG. 3 is a signaling diagram showing an example of authentication.

[0010] FIG. 4 is a flow chart showing an example method for authentication.

[0011] FIG. 5 is a flow chart showing an example method for encoding data.

[0012] FIG. 6 is a flow chart showing an example method for decoding data.

DETAILED DESCRIPTION

[0013] The accompanying drawings, which form a part hereof, show examples of the disclosure.

[0014] It is to be understood that the examples shown in the drawings and/or discussed herein are non-exclusive and that there are other examples of how the disclosure may be practiced.

[0015] FIG. 1 shows an example communication network 100 in which features described herein may be implemented. The communication network 100 may comprise one or more information distribution networks of any type, such as, without limitation, a telephone network, a wireless network (e.g., an LTE network, a 5G network, a WiFi IEEE 802.11 network, a WiMAX network, a satellite network, and/or any other network for wireless communication), an optical fiber network, a coaxial cable network, and/or a hybrid fiber/coax distribution network. The communication network 100 may use a series of interconnected communication links 101 (e.g., coaxial cables, optical fibers, wireless links, etc.) to connect multiple premises 102 (e.g., businesses, homes, consumer dwellings, train stations, airports, etc.) to a local office 103 (e.g., a headend). The local office 103 may send downstream information signals and receive upstream information signals via the communication links 101. Each of the premises 102 may comprise devices, described below, to receive, send, and/or otherwise process those signals and information contained therein.

[0016] The communication links 101 may originate from the local office 103 and may comprise components not illustrated, such as splitters, filters, amplifiers, etc., to help convey signals clearly. The communication links 101 may be coupled to one or more wireless access points 127 configured to communicate with one or more mobile devices 125 via one or more wireless networks. The mobile devices 125 may comprise smart phones, tablets or laptop computers with wireless transceivers, tablets or laptop computers communicatively coupled to other devices with wireless transceivers, and/or any other type of device configured to communicate via a wireless network.

[0017] The local office 103 may comprise an interface 104, such as a termination system (TS). The interface 104

may comprise a cable modem termination system (CMTS) and/or other computing device(s) configured to send information downstream to, and to receive information upstream from, devices communicating with the local office **103** via the communications links **101**. The interface **104** may be configured to manage communications among those devices, to manage communications between those devices and backend devices such as servers **105-107** and **122**, and/or to manage communications between those devices and one or more external networks **109**. The local office **103** may comprise one or more network interfaces **108** that comprise circuitry needed to communicate via the external networks **109**. The external networks **109** may comprise networks of Internet devices, telephone networks, wireless networks, wireless networks, fiber optic networks, and/or any other desired network. The local office **103** may also or alternatively communicate with the mobile devices **125** via the interface **108** and one or more of the external networks **109**, e.g., via one or more of the wireless access points **127**.

[0018] The intermediate device **122** may intercept incoming communications and process those communications to be forwarded to a variety of computing devices, such as push server **105**, content server **106**, app server **107**, and other computing devices. The intermediate device **122** may receive communications between devices located in a local office **103** and devices outside the local office **103** communicating via communication links **101**. The intermediate device **122** may terminate secure communication channels established over communication links **101**, such as SSL and TLS connections. The intermediate device **122** may include any of a variety of devices, such as load balancers, internet gateways, network address translation (NAT) gateways, routes, and the like. The push notification server **105** may be configured to generate push notifications to deliver information to devices in the premises **102** and/or to the mobile devices **125**. The content server **106** may be configured to provide content to devices in the premises **102** and/or to the mobile devices **125**. This content may comprise, for example, video, audio, text, web pages, images, files, etc. The content server **106** (or, alternatively, an authentication server) may comprise software to validate user identities and entitlements, to locate and retrieve requested content, and/or to initiate delivery (e.g., streaming) of the content. The application server **107** may be configured to offer any desired service. For example, an application server may be responsible for collecting, and generating a download of, information for electronic program guide listings. Another application server may be responsible for monitoring user viewing habits and collecting information from that monitoring for use in selecting advertisements. Yet another application server may be responsible for formatting and inserting advertisements in a video stream being transmitted to devices in the premises **102** and/or to the mobile devices **125**. The local office **103** may comprise additional servers, such as the intermediate device **122**, additional push, content, and/or application servers, and/or other types of servers. Although shown separately, the push server **105**, the content server **106**, the application server **107**, the intermediate device **122**, and/or other server(s) may be combined. The servers **105**, **106**, **107**, and **122**, and/or other servers, may be computing devices and may comprise memory storing data and also storing computer executable instructions that, when executed by one or more processors, cause the server(s) to perform steps described herein.

[0019] An example premises **102a** may comprise an interface **120**. The interface **120** may comprise circuitry used to communicate via the communication links **101**. The interface **120** may comprise a modem **110**, which may comprise transmitters and receivers used to communicate via the communication links **101** with the local office **103**. The modem **110** may comprise, for example, a coaxial cable modem (for coaxial cable lines of the communication links **101**), a fiber interface node (for fiber optic lines of the communication links **101**), twisted-pair telephone modem, a wireless transceiver, and/or any other desired modem device. One modem is shown in FIG. 1, but a plurality of modems operating in parallel may be implemented within the interface **120**. The interface **120** may comprise a gateway **111**. The modem **110** may be connected to, or be a part of, the gateway **111**. The gateway **111** may be a computing device that communicates with the modem(s) **110** to allow one or more other devices in the premises **102a** to communicate with the local office **103** and/or with other devices beyond the local office **103** (e.g., via the local office **103** and the external network(s) **109**). The gateway **111** may comprise a set-top box (STB), digital video recorder (DVR), a digital transport adapter (DTA), a computer server, and/or any other desired computing device.

[0020] The gateway **111** may also comprise one or more local network interfaces to communicate, via one or more local networks, with devices in the premises **102a**. Such devices may comprise, e.g., display devices **112** (e.g., televisions), STBs or DVRs **113**, personal computers **114**, laptop computers **115**, wireless devices **116** (e.g., wireless routers, wireless laptops, notebooks, tablets and netbooks, cordless phones (e.g., Digital Enhanced Cordless Telephone-DECT phones), mobile phones, mobile televisions, personal digital assistants (PDA)), landline phones **117** (e.g., Voice over Internet Protocol-VoIP phones), and any other desired devices. Example types of local networks comprise Multimedia Over Coax Alliance (MoCA) networks, Ethernet networks, networks communicating via Universal Serial Bus (USB) interfaces, wireless networks (e.g., IEEE 802.11, IEEE 802.15, Bluetooth), networks communicating via in-premises power lines, and others. The lines connecting the interface **120** with the other devices in the premises **102a** may represent wired or wireless connections, as may be appropriate for the type of local network used. One or more of the devices at the premises **102a** may be configured to provide wireless communications channels (e.g., IEEE 802.11 channels) to communicate with one or more of the mobile devices **125**, which may be on- or off-premises.

[0021] The mobile devices **125**, one or more of the devices in the premises **102a**, and/or other devices may receive, store, output, and/or otherwise use assets. An asset may comprise a video, a game, one or more images, software, audio, text, webpage(s), and/or other content.

[0022] FIG. 2 shows hardware elements of a computing device **200** that may be used to implement any of the computing devices shown in FIG. 1 (e.g., the mobile devices **125**, any of the devices shown in the premises **102a**, any of the devices shown in the local office **103**, any of the wireless access points **127**, any devices with the external network **109**) and any other computing devices discussed herein. The computing device **200** may comprise one or more processors **201**, which may execute instructions of a computer program to perform any of the functions described herein. The instructions may be stored in a read-only memory (ROM)

202, random access memory (RAM) **203**, removable media **204** (e.g., a USB drive, a compact disk (CD), a digital versatile disk (DVD)), and/or in any other type of computer-readable medium or memory. Instructions may also be stored in an attached (or internal) hard drive **205** or other types of storage media. The computing device **200** may comprise one or more output devices, such as a display device **206** (e.g., an external television and/or other external or internal display device) and a speaker **214**, and may comprise one or more output device controllers **207**, such as a video processor. One or more user input devices **208** may comprise a remote control, a keyboard, a mouse, a touch screen (which may be integrated with the display device **206**), microphone, etc. The computing device **200** may also comprise one or more network interfaces, such as a network input/output (I/O) interface **210** (e.g., a network card) to communicate with an external network **209**. The network I/O interface **210** may be a wired interface (e.g., electrical, RF (via coax), optical (via fiber)), a wireless interface, or a combination of the two. The network I/O interface **210** may comprise a modem configured to communicate via the external network **209**. The external network **209** may comprise the communication links **101** discussed above, the external network **109**, an in-home network, a network provider's wireless, coaxial, fiber, or hybrid fiber/coaxial distribution system (e.g., a DOCSIS network), or any other desired network. The computing device **200** may comprise a location-detecting device, such as a global positioning system (GPS) microprocessor **211**, which may be configured to receive and process global positioning signals and determine, with possible assistance from an external server and antenna, a geographic position of the computing device **200**.

[0023] Although FIG. 2 shows an example hardware configuration, one or more of the elements of the computing device **200** may be implemented as software or a combination of hardware and software. Modifications may be made to add, remove, combine, divide, etc. components of the computing device **200**. Additionally, the elements shown in FIG. 2 may be implemented using basic computing devices and components that have been configured to perform operations such as are described herein. For example, a memory of the computing device **200** may store computer-executable instructions that, when executed by the processor **201** and/or one or more other processors of the computing device **200**, cause the computing device **200** to perform one, some, or all of the operations described herein. Such memory and processor(s) may also or alternatively be implemented through one or more Integrated Circuits (ICs). An IC may be, for example, a microprocessor that accesses programming instructions or other data stored in a ROM and/or hardwired into the IC. For example, an IC may comprise an Application Specific Integrated Circuit (ASIC) having gates and/or other logic dedicated to the calculations and other operations described herein. An IC may perform some operations based on execution of programming instructions read from ROM or RAM, with other operations hardwired into gates or other logic. Further, an IC may be configured to output image data to a display buffer.

[0024] Transport Layer Security (TLS) is a standard security protocol (RFC 5246) providing a cryptographic solution for a client (such as a web browser) to authenticate a target device. TLS may also include establishing an encrypted secure channel via which the sending device and target device can exchange information. Most implementations of

TLS are referred to as 1-way TLS in which computing devices are authenticated using a certificate during a handshaking procedure. It should be noted that TLS is provided as an example only and that any security protocol (or combination of protocols) may be used as appropriate. Computing devices may be authenticated via any authentication mechanism, such as a user ID and password. A sending device attempting to access a target device via a security protocol may directly communicate with an intermediate device, such as a gateway or load balancer, rather than with the target device directly. This intermediate device may serve as the endpoint for the communications being received from the sending device and may forward the received communications to the target device via a separate communication channel. This termination of the secured communication (e.g. TLS termination) may include handling incoming secured connections, decrypting the secured data, and passing on the unencrypted request to the target device as it may be assumed that the connection between the intermediate device and the target device is secure. That is, security measures, undertaken by computing devices to secure communications between the computing device and a target device, may only extend to an intermediate device rather than the target device. The computing device relies on the intermediate device to ensure that communications between the computing device and the target device are secure. However, a sending device may distrust the connection between an intermediate device and a target device is secure. Additionally, the sending device may distrust the intermediate device as the intermediate device may be able to read and/or modify the data intended for the target device.

[0025] Many devices in machine-to-machine systems, such as internet-of-things (IoT) devices, encounter issues when handshaking information to establish a secure communication channel is intercepted by intermediate devices (e.g. TLS termination), such as load balancers, gateway devices, and other high availability infrastructure. The devices and the target device, which are communicating data with each other, are therefore unable to complete the handshake process as the target device does not participate in security authentication and relies on information received from the intermediate device. Accordingly, the data is not transmitted by a known secure communication channel from endpoint to endpoint, resulting in communications that are not guaranteed to be secure from endpoint to endpoint. In order to verify the security of the communication channel from the sending device to the target device, security data (e.g., a security certificate for authentication) may be structured at the application layer such that it passes through the intermediate device and from the sending device and/or target device without the security data being extracted or processed at the intermediate device. The security data may be encoded and presented at an application layer rather than at a link layer (typically used by tunneling applications) as in the Internet protocol suite. For example, a sending device may send secured data to a cloud service such that the secured data passes through a gateway and to a destination application behind the gateway.

[0026] FIG. 3 is a diagram showing an example authentication process. At step **310**, data may be assembled by a sending device. The data may specify the highest security protocol version it supports, a random number, a list of suggested cipher suites, and/or compression methods. As described in more detail with respect to FIG. 5, the message

may include an encoded copy of the sending device's public key certificate. The sending device may be any computing device that transmits data intended for the target device. For example, the sending device may be a device managed by a first content provider, and the intermediate device and/or target device may be managed by a second content provider. In this example, the sending device may wish to transmit data associated with the first content provider to the target device without exposing the data to the second content provider as the data passes through the intermediate device. For example, a first service provider may manage both the sending device and the target device and a second service provider may manage the intermediate device. The first service provider may wish to transmit data from the sending device to the target device without exposing the transmitted data to the second service provider as the data passes through the intermediate device. For example, sending devices may generate and/or send a variety of data regarding an environment to a central server for processing and analysis, such as in smart home applications for reducing energy usage and/or monitoring occupant behaviors. For example, sending devices can be used for remote health monitoring and emergency notification systems. Health monitoring devices may include blood pressure monitors, heart rate monitors, activity trackers, pacemakers, hearing aids, or any other device monitoring and/or controlling the health of an individual. Data generated from these devices may be sent to a target device at a hospital, insurance company, and/or any other treatment facility to monitor the health and wellbeing of a patient in a secure manner.

[0027] At step 312, a secure communication channel may be requested. The secure communication channel may be a security request (such as a TLS request) for the target device. The request may include the data assembled by the sending device. The intermediate device may prepare a message indicating the chosen protocol version, a random number, cipher suite, a compression method from the choices offered by the sending device, a session identifier, and/or a public key associated with the intermediate device and/or a public key associated with the target device. The chosen protocol version may be the highest security protocol version that both the sending device and intermediate device support. The secure communication channel may be requested so that data transmitted between the sending device and the target device is secured from analysis or modification by third parties. For example, a first content provider may secure communications between a sending device managed by the first content provider and a target device managed by the second content provider so that the communications between the sending device and the target device cannot be analyzed or modified by the second content provider. For example, a sending device and a target device, both managed by a first service provider, may secure communications between the sending device and the target device so that a second service provider, which may manage an intermediate device routing communications between the sending device and the target device, cannot analyze or modify the communications sent between the sending device and the target device. For example, a smart thermostat may request a secured communication channel with a central server for transmitting temperature data over a particular time period. For example, a health monitoring device may request a

secure communication channel with a target device at a hospital so that a doctor can monitor the vital signs of a patient.

[0028] At step 314, the intermediate device may establish a secure communication channel between the intermediate device and the sending device. The intermediate device may transmit the prepared message to the sending device. The sending device may respond with a message indicating the public key of the sending device and encrypted using the public key of the intermediate device and/or the target device. The sending device and intermediate device may compute a common encryption key that may be used to encrypt communications between the sending device and the intermediate device. The public keys and/or common encryption key may be used to establish a secure communication channel between the sending device and the intermediate device. The secure communication channel may include a Hypertext Transfer Protocol (HTTP) or HTTP Secure (HTTPS) session. The protocol version used in a HTTPS session may correspond to the chosen protocol version established in the message prepared by the intermediate device. The secure communication channel may be encrypted using the public key associated with the sending device, intermediate device, and/or target device. The secure communication channel may be encrypted using the common encryption key. For example, an intermediate device managed by a second content provider may act as a gateway for communications sent to a target device managed by the second content provider. The intermediate device may intercept communications intended for the target device. For example, an intermediate device managed by a second service provider may provide a routing service for communications between one or more sending devices and a target device provided by a first service provider. The intermediate device may route communications from a variety of sending devices to the target device via a dedicated, secured communication channel between the intermediate device and the target device. For example, a smart thermostat may establish a secure communication channel with a load balancer used by a central server system for monitoring the environmental conditions across multiple homes. For example, a health monitoring devices may establish a secure communication channel with a gateway device that secures data stored by the target device from access by devices outside of the hospital.

[0029] At step 316, the sending device may send data to the intermediate device via the secure communication channel. The data sent by the sending device may be intended for the target device. The intermediate device may serve as the endpoint of the secure communication channel intended between the sending device and the target device (e.g. TLS termination). The data sent by the sending device include a header and a body. As described in more detail with respect to FIG. 5, the header may include a variety of data, including a copy of the sending device's public key certificate. The body may include an encoded copy of the sending device's public key certificate and/or any data, such as data generated by an IoT device, to be sent to the target device. For example, the header and/or body may be encrypted by a sending device managed by a first content provider and sent to a target device managed by a second content provider so that an intermediate device provided by the second content provider cannot analyze or modify the encrypted data. For example, the header and/or body may be encrypted by a

sending device managed by a first service provider and sent to a target device managed by the first service provider so that an intermediate device managed by a second service provider cannot decode the encrypted data. For example, a smart thermostat may send data packets including the temperature data intended for a central server to a load balancer. For example, a health monitoring device may send private health data, intended for a doctor's computer, that is received by a gateway device.

[0030] The intermediate device may send, at step 318, the data provided by the sending device to the target device. For example, the intermediate device may provide a gateway service and route incoming communications to a variety of computing devices such as the target device. For example, the encrypted data in the header and/or body may be sent by the intermediate device managed by a second content provider to the target device managed by the second content provider so that the intermediate device cannot analyze or modify the encrypted data sent by the first content provider. For example, the encrypted data in the header and/or body may be sent by an intermediate device managed by a second service provider to a target device managed by a first service provider so that the intermediate device cannot process the encrypted data sent by the target device. The intermediate device may decode the data provided by the sending device without modifying the header and/or body of the data sent by the sending device. For example, a portion of the header and/or body may be encrypted using a public key provided by the target device and/or a private key used by the sending device such that the encrypted portion of the header and/or body cannot be readily viewed or processed by the intermediate device. The intermediate device may decode the data using the public key associated with the sending device, intermediate device, and/or target device. The intermediate device may decode the data using a common encryption key. The data sent by the intermediate device to the target device may be sent via a second communication channel distinct from the secure communication channel established by the sending device and intermediate device. For example, the target device may store data captured by a variety of IoT device and the intermediate device may be a load balancing proxy handling requests to upload captured data. The load balancing proxy and the target device may have a previously established second communication channel between them. The second communication channel may be unsecured. The second communication channel may be secured using self-signed certificates issued by the target device and/or intermediate device. For example, a load balancer may select a target device from a plurality of target devices providing a distributed central server and transmit the received data to the selected target device. For example, a gateway may forward incoming data to a target device associated with a patient's doctor without exposing the target device to any public networks, such as the Internet.

[0031] At step 320, the identity of the transmitting device (or any other sending device) may be validated by the target device. The target device may reassemble the data sent by the sending device. The data may be encoded and the target device may decode the encoded request data. Certificate data may be extracted, by the target device, from the decoded request data. The certificate data may be decoded using a public key associated with the sending device. The certificate data may be validated by a validation system and/or by a public key provided by the sending device. The validation

of the certificate data may include verifying the certificate is not expired and/or the certificate was issued by a trusted validation system. The target device may validate signature data based on the verified certificate. The target device may validate the identity of the sending device by providing the sending device's public key certificate to a validation system. Specific validation techniques are described in more detail with respect to FIG. 6. For example, a smart thermostat may encode temperature data using a certificate associated with the smart thermostat and a central server and the central server can verify the identity of the smart thermostat using the certificate. For example, a doctor's computer can validate that health data received from a patient's health monitoring device was generated by the health monitoring device.

[0032] At step 322, the received data may be processed by the target device. Any of a variety of data may be received and processed by the target device. For example, a target device may receive data regarding an environment from a smart thermostat installed in a user's home. For example, a target device may receive health information from a variety of health monitoring devices, such as blood pressure monitors, heart rate monitors, activity trackers, pacemakers, hearing aids, or any other device monitoring and/or controlling the health of an individual. A target device at a hospital, insurance company, and/or any other treatment facility may monitor the health and wellbeing of a patient in a secure manner based on the received data. For example, sending devices may include location-tracking capabilities and be attached to any of a variety of vehicles, cargo, and other vehicular infrastructure devices. These devices may capture data describing the location and/or operation of a variety of vehicles and other objects and be provided to a target device in a secure manner. The target device may process the vehicular data for smart traffic control, electronic toll systems, vehicle telematics, fleet management, vehicle control, and/or providing roadside assistance.

[0033] At step 324, the target device may send data to the sending device. The data may be sent directly from the target device to the sending device. The target device may provide the data to the intermediate device and the data may be sent to the sending device via the secure communication channel provided between the intermediate device and the sending device. The data sent by the target device may be based on the data sent by the sending device. For example, the target device may determine a thermostat setting and transmit the thermostat setting to a smart thermostat based on energy usage and temperature data provided by the smart thermostat. For example, the target device may transmit a notification message to an activity tracker indicating that a blood pressure measurement provided by the activity tracker exceeds a threshold value determined by the target device. For example, the target device may transmit routing data to a vehicle telematics device when the target device determines that the location of a vehicle telematics device has deviated from a route previously provided to the vehicle telematics device.

[0034] A sending device may format data to be sent, via a secure communication channel, to a target device such that security information, such as the sending device's public key certificate, will be provided to a target device even in the event of an intermediate device terminating the secure communication channel. FIG. 4 is a flow chart showing an example method for authentication, such as TLS authenti-

cation. Some or all of the steps described with respect to FIG. 4 may be performed by a sending device, a target device, and/or any other computing device.

[0035] Certificate data may be received (410). Computing devices may utilize public certificates to validate the sending device and/or target device. Each device has their own public certificate and may authenticate other devices via a handshaking procedure. These certificates may utilize a public key infrastructure (PKI) in which a certificate authority, or other computing device, issues certificates to particular devices and validates a provided certificate. The validation system may provide a public key associated with a sending device that may be retrieved when validating a certificate. In this way, the validation system may validate that data signed with the certificate for a particular system originated from the particular system. However, distributed authentication systems (such as OpenPGP) may also be utilized to verify the authenticity of a certificate. Many certificates are formatted using the X.509 standard. The certificate data may include a X.509 certificate. However, any digital certificate or digital signature generated using any cryptosystem, such as RSA, DSS, elliptical curves, and YAK, may be utilized. The certificate data may be provided by a validation system.

[0036] A digital signature may be generated (412). The digital signature may be generated by a sending device. The digital signature may be generated based on the sending device's public key. The digital signature may be generated using a Base64 encoding. However, any encoding, such as MIME and yEnc, may also be utilized. The digital signature may also include a signature for the sending device. The signature of the sending device may be generated based on any of a variety of hardware and/or software aspects of the sending device. Hardware and/or software aspects of the sending device may include, but are not limited to, a media access control address, IP address, installed processor(s), memory, operating system, and/or installed software. The signature may be generated using any of a variety of cryptographic hash functions, such as SHA-2.

[0037] Request data may be generated (414). The request data may include the encoded digital signature as described in more detail with respect to FIG. 5. The request data may include a header and/or a body. The request data may be a message formatted using HTTP. The request data may be a HTTP request and/or a HTTP response. The header may include a variety of data, including the encoded digital signature. The body may include the encoded digital signature, and/or any data, such as data generated by an IoT device, to be sent to the target device. For example, a smart home device may generate data describing the time a HVAC system is in operation. For example, a health monitoring device may generate data describing a patient's heartbeat. For example, a vehicle telematics device may generate data regarding the location of the vehicle telematics device and the speed at which the vehicle telematics device is moving.

[0038] Request data may be sent (416). The request data may be sent, by the sending device and via a communication channel, and intended for a target device. The request data may be intercepted by an intermediate device that forwards the header data to the target device. The communication channel may be secured between the sending device and an intermediate device. Data sent via the secure communication channel may be encrypted using a public key associated with the sending device, intermediate device, and/or target

device. The secure communication channel may be encrypted using a common encryption key negotiated between the sending device and the intermediate device. The request data may include the encoded data in the header. The header may be a HTTP and/or HTTPS header. For example, a smart home device may send data to a central server which provides the data to a homeowner's mobile device. For example, a health monitoring device may send data to a hospital server system so that the patient's doctor can keep track of the patient's condition. For example, a vehicle telematics device may send location to a fleet tracking system which tracks the location of a fleet of vehicles.

[0039] Request data may be received (418). The request data may be received by a target device and from an intermediate device. The intermediate device may decrypt the request data using the security data for the secure communication channel between the sending device and the intermediate device. The intermediate device may decode the data using the public key associated with the sending device, intermediate device, and/or target device and/or the common encryption key. The data sent by the intermediate device to the target device may be sent via a second communication channel distinct from the secure communication channel established by the sending device and intermediate device. The second communication channel may be unsecured. The second communication channel may be secured using self-signed certificates issued by the target device and/or intermediate device.

[0040] A digital signature may be decoded (420). The target device may decode the request data to receive the digital signature for the sending device originally sending the request data. The digital signature may be decoded by the target device as described in more detail with respect to FIG. 6.

[0041] Request data may be validated (422). The validation of the request data may be based on a request provided by the target device to a validation system. The target device may validate the identity of the originator of the request data using the digital signature. The request data may be validated based on certificate data validated by the validation system that provided the certificate data associated with the sending device.

[0042] FIG. 5 is a flow chart showing an example method for encoding data. Some or all of the steps described with respect to FIG. 5 may be performed by a sending device and/or any other computing device.

[0043] Certificate data may be received (510). The certificate data may be generated by the sending device. The certificate data may be generated by a validation system for the sending device. The certificate data may include a public key infrastructure (PKI) certificate. The validation system may be provided by a certificate authority that issues certificates to particular devices and validates a provided certificate. A certificate authority may be a neutral, trusted third party that is trusted by both the sending device and the target device. The certificate may be a root certificate and/or an intermediate certificate. A root certificate may be utilized by a computing device to issue an intermediate certificate.

[0044] Timestamp data may be received (512). The timestamp may indicate the time the request was generated. The timestamp may be generated using the internal clock of the sending device. The timestamp may be received from a time service, such as a network time protocol server. The time-

stamp may be based on a message indicating a time received by the sending device and from the intermediate device and/or the target device.

[0045] A type of the certificate may be determined (514). The certificate type may be determined by comparing the format of the digital certificate to known certificate formats. The certificate type may be indicated by metadata associated with the certificate. The certificate type may be indicated by the header of the request. The certificate data may include a X.509 certificate. However, any digital certificate or digital signature generated using any cryptosystem, such as RSA, DSS, elliptical curves, and YAK, may be utilized. The certificate data may be provided by a validation system. Distributed authentication systems (such as OpenPGP) may also be utilized to verify the authenticity of a certificate.

[0046] A target device may be determined (516). The target device may be identified by an IP address, domain name, or any other identifier. The identity of the target device may be pre-coded on the sending device. For example, the sending device may be programmed with the URL of a target device that stores data being captured by the sending device. However, any identification, such as an IP address, may be used to identify the target device. The identity of the target device may be based on the data being sent to the target device. For example, the sending device may capture two distinct types of data, with the first type of data being sent to a first target device and a second type of data being sent to a second target device.

[0047] Nonce data may be generated (518). The nonce data may include any random or pseudo-random number. The nonce data may be generated by the sending device. The nonce may be valid for a single use. The nonce data may include a timestamp and/or may be time-variant. The nonce data may be generated with a sufficient length as to be probabilistically unlikely to repeat a previously generated nonce value.

[0048] A request certificate may be generated (520). The request certificate may include the certificate data, the time-stamp, the target device, an indication of the certificate type, and/or the nonce. The following is an example data structure for the request certificate:

```

"certificateRequestToken": {
  "certificate": "<certificate data>",
  "timestamp": 1503687297540,
  "type": "x509"
  "target": "server.host.com"
  "nonce": "1JNiy1Zc9to7BwDRfR"
}

```

[0049] Request data may be encoded (522). The request data may include the request token and a digital signature. The digital signature may be encoded by concatenating the raw byte values of the request certificate and encoding the concatenated byte values. The digital signature may be encoded using any of a variety of digital signatures, such as a private key on the sending device. The following is an example data structure for the request data:

```

"certificateRequestToken": {
  "certificate": "<certificate data>",
  "timestamp": 1503687297540,
  "type": "x509"
}

```

-continued

```

"target": "server.host.com"
"nonce": "1JNiy1Zc9to7BwDRfR"
"signature": "<digital signature>"
}

```

[0050] The request data may be included in data to be sent from a sending device to the target device. The data to be sent may include a header and/or a body. The request data may be included in the header and/or the body. The request data may be encoded using any appropriate encoding scheme based on the communication channel, such as by computing a Base64 string. The following is an example of a header having encoded request data:

```

POST https://www.host.com/preferenceDataService/
Connection: keep-alive
Accept: application/json
Content-Type: application/json
User-Agent: YourUserAgent
X-PartnerId: cox
Authorization: Bearer <encoded request data>

```

[0051] A target device may receive request data, decode the received request data, and confirm the identity of a sending device sending the request data using a digital signature in the received request data. FIG. 6 is a flow chart showing an example method for decoding data. Some or all of the steps described with respect to FIG. 6 may be performed by a target device and/or any other computing device.

[0052] Request data may be received (610). The request data may be sent by and received from an intermediate device. The request data may originate from a sending device. The request data may be encoded using security data for a communication channel between the intermediate device and the target device. The request data may be decoded using the security information for the communication channel between the intermediate device and the target device. The request data may include a header and/or a body. The header and/or body may include the encoded digital signature and/or any data, such as data generated by an IoT device, to be sent to the target device.

[0053] Request data may be reassembled (612). As sent by the sending device, the request data may be encoded using any of a variety of encoding formats, such as Base64, MIME, and/or yENC. Reassembling the request data may include decoding the encoded request data. Reassembling the request data may include performing a Base64 decode on the request data. However, any decoding may be performed on the request data, such as a MIME decode and/or a yEnc decode. The request data may be reassembled in order to convert the request data from a byte string into a plain text format. For example, the reassembly of the request data may include performing a Base64 decode on a byte string to generate a HTTP request message (and/or a HTTP response message) which may be decoded by a web server.

[0054] Certificate data may be extracted (614). The certificate data may be extracted from the reassembled request data. The certificate data may be extracted from a header and/or a body in the reassembled request data. For example, the certificate data may be extracted from a HTTP header in the reassembled request data. Extracting the certificate data

may include decoding the certificate data. The certificate data may be decoded using a public key associated with the sending device. The sending device may provide the public key. A target device may receive the public key from a validation system such as a certificate authority. The public key may be utilized to decode data encoded by a sending device to verify that the data was sent by the sending device.

[0055] Certificate data may be validated (616). The certificate data may be validated by a validation system. The certificate data may be validated by a public key provided by the sending device. The validation of the certificate data may include verifying the certificate is valid. The validation of the certificate data may include verifying the certificate was issued by a trusted validation system. The validation of the certificate data may include verifying that the type of certificate identified in the request data matches the type of certificate extracted from the request data. For example, if the header of the request data identifies the certificate as a X.509 certificate, the extracted certificate may only be valid if the extracted certificate is a X.509 certificate.

[0056] Signature data may be validated (618). The signature data may be validated to confirm the device encoding the signature data corresponds to the device sending the data. The signature data may be validated using a public key provided by the validation system. The signature data may be validated using a public key provided by the sending device. The signature data may be validated using the certificate data once it has been validated. The signature data may be validated by a public key contained in the certificate data. The signature data may be validated using a signature generated based on the hardware and/or software configuration of the sending device. The signature may be validated by decoding the signature data and comparing the signature data to the request data with the signature removed. For example, the decoded signature data may correspond to the request data without the header such that an identical header should be decoded from the signature data. In this way, it may be confirmed that the computing device generating the request data also generated the signature. This comparison may be done in any of a variety of ways, such as character-by-character, field-by-field, etc. For example, the decoded header and the header in the request data may each be interpreted as a string of characters and the comparison may be done by comparing each character in the decoded header string to each character in the request header string. For example, each field in the decoded header (certificate, timestamp, type, target, nonce, and/or etc.) may be compared to the corresponding field in the header of the request data.

[0057] The request data may be validated (620). The request data may be validated by comparing a timestamp in the request data to the time at which the request data was received. The receipt time may be determined based on a current time maintained by the target device. If the request data was received outside a threshold period of time, the request data may be considered invalid. In this way, the target device can ensure that previously valid request data provided by the sending device is not being replayed by a malicious actor. The request data may be validated by verifying the integrity of the nonce value. The integrity of the nonce value may be determined by comparing the nonce value to a database of nonce values previously received by the target device and/or from the sending device. If the nonce value has been previously received by the target device from the sending device (either at any time in the past

or within a threshold period of time, such as five minutes), the request data may be considered to be invalid. In this way, the target device can ensure that previously valid request data provided by the sending device has not been modified by a malicious actor.

[0058] Although examples are described above, features and/or steps of those examples may be combined, divided, omitted, rearranged, revised, and/or augmented in any desired manner. Various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of this description, though not expressly stated herein, and are intended to be within the spirit and scope of the disclosure. Accordingly, the foregoing description is by way of example only, and is not limiting.

1. A method comprising:
 - generating, by a first device, a data package comprising:
 - certificate data associated with the first device, and
 - an encrypted version of the certificate data; and
 - establishing, based on the certificate data in the data package, a secure communication channel between the first device and an intermediate device, wherein the intermediate device is blocked from decrypting the encrypted version of the certificate data.
2. The method of claim 1, wherein the generating the data package further comprises:
 - generating a nonce;
 - generating a timestamp;
 - generating an indication of a type of the certificate data; and
 - including the nonce, the timestamp, and the indication of the type in the data package.
3. The method of claim 1, further comprising:
 - encrypting the data package using a private key associated with the first device.
4. The method of claim 1, wherein generating the encrypted version of the certificate data comprises:
 - encrypting the certificate data using a public key associated with a second device.
5. The method of claim 1, further comprising:
 - sending, by the first device, to a second device, and via the intermediate device, data that is encrypted by the first device;
 - wherein the first device and the second device are managed by a first service provider, and the intermediate device is managed by a second service provider.
6. A method comprising:
 - receiving, by a second device and from an intermediate device, a data package comprising:
 - certificate data, and
 - an encrypted version of the certificate data;
 - validating the data package based on the certificate data and based on decrypting the encrypted version of the certificate data using a private key associated with the second device; and
 - verifying an identity of a first device based on the certificate data.
7. The method of claim 6, wherein the intermediate device is blocked from decrypting the encrypted version of the certificate data included in the data package.
8. The method of claim 6, further comprising decrypting the data package using a public key associated with the first device.

9. The method of claim 6, further comprising performing a textual decoding operation on the data package to obtain binary data.

10. The method of claim 6, wherein the validating the data package further comprises:

comparing, by the second device:

- a nonce value indicated in the data package; and
- a database of previously received nonce values.

11. A first device comprising:

one or more processors; and

memory storing instructions that, when executed by the one or more processors, cause the first device to:

generate a data package comprising:

- certificate data associated with the first device, and
- an encrypted version of the certificate data; and
- establish, based on the certificate data in the data package, a secure communication channel between the first device and an intermediate device,

wherein the intermediate device is blocked from decrypting the encrypted version of the certificate data.

12. The first device of claim 11, wherein the instructions, when executed by the one or more processors, further cause the first device to:

- generate a nonce;
- generate a timestamp;
- generate an indication of a type of the certificate data; and
- include the nonce, the timestamp, and the indication of the type in the data package.

13. The first device of claim 11, wherein the instructions, when executed by the one or more processors, further cause the first device to:

encrypt the data package using a private key associated with the first device.

14. The first device of claim 11, wherein the instructions, when executed by the one or more processors, further cause the first device to:

encrypt the certificate data using a public key associated with a second device to generate the encrypted version of the certificate data.

15. The first device of claim 11, wherein the instructions, when executed by the one or more processors, further cause the first device to:

send, to a second device and via the intermediate device, data that is encrypted by the first device;

wherein the first device and the second device are managed by a first service provider, and the intermediate device is managed by a second service provider.

16. A second device comprising:

one or more processors; and

memory storing instructions that, when executed by the one or more processors, cause the second device to:

receive, from an intermediate device, a data package comprising:

- certificate data; and
- an encrypted version of the certificate data;

validate the data package based on the certificate data and based on decrypting the encrypted version of the certificate data using a private key associated with the second device; and

verify an identity of a first device based on the certificate data.

17. The second device of claim 16, wherein the instructions, when executed by the one or more processors, further cause the second device to:

operate in a context where the intermediate device is blocked from decrypting the encrypted version of the certificate data included in the data package.

18. The second device of claim 16, wherein the instructions, when executed by the one or more processors, further cause the second device to:

decrypt the data package using a public key associated with the first device.

19. The second device of claim 16, wherein the instructions, when executed by the one or more processors, further cause the second device to:

perform a textual decoding operation on the data package to obtain binary data.

20. The second device of claim 16, wherein the instructions, when executed by the one or more processors, further cause the second device to:

compare a nonce value indicated in the data package to a database of previously received nonce values as part of validating the data package.

21. A non-transitory computer-readable medium storing instructions that, when executed, cause:

generating, by a first device, a data package comprising: certificate data associated with the first device, and an encrypted version of the certificate data; and establishing, based on the certificate data in the data package, a secure communication channel between the first device and an intermediate device, wherein the intermediate device is blocked from decrypting the encrypted version of the certificate data.

22. The non-transitory computer-readable medium of claim 21, wherein the instructions, when executed, further cause:

generating a nonce;

generating a timestamp;

generating an indication of a type of the certificate data; and

including the nonce, the timestamp, and the indication of the type in the data package.

23. The non-transitory computer-readable medium of claim 21, wherein the instructions, when executed, further cause:

encrypting the data package using a private key associated with the first device.

24. A non-transitory computer-readable medium storing instructions that, when executed, cause:

receiving, by a second device and from an intermediate device, a data package comprising:

certificate data, and

an encrypted version of the certificate data;

validating the data package based on the certificate data and based on decrypting the encrypted version of the certificate data using a private key associated with the second device; and

verifying an identity of a first device based on the certificate data.

25. The non-transitory computer-readable medium of claim 24, wherein the instructions, when executed, further cause:

operating in a context in which the intermediate device is blocked from decrypting the encrypted version of the certificate data included in the data package.

26. The non-transitory computer-readable medium of claim **24**, wherein the instructions, when executed, further cause:

performing a textual decoding operation on the data package to obtain binary data.

27. A system comprising:

a first device;

an intermediate device; and

a second device,

wherein the first device is configured to:

generate a data package comprising:

certificate data associated with the first device, and

an encrypted version of the certificate data; and

establish, based on the certificate data in the data package, a secure communication channel between the first device and the intermediate device,

wherein the intermediate device is blocked from decrypting the encrypted version of the certificate data, and

wherein the second device is configured to receive, from the intermediate device, the data package.

28. The system of claim **27**, wherein the first device is further configured to:

send, via the intermediate device and to the second device, the data package.

29. The system of claim **27**, wherein the second device is further configured to:

validate the data package based on the certificate data and based on decrypting the encrypted version of the certificate data using a private key associated with the second device.

30. The system of claim **27**, wherein the second device is further configured to:

perform a textual decoding operation on the data package to obtain binary data.

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